COXIS-high tumor nests and LRRC15 CAF stroma form therapy-resistant niches in breast cancer

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INTRODUCTION

Breast cancer is a biologically and clinically heterogeneous disease in which therapeutic resistance remains a major cause of treatment failure. Although immune checkpoint blockade (ICB) has produced durable responses in some cancers, the majority of breast tumors do not respond, particularly those with immune-excluded or "cold" tumor microenvironments. Increasing evidence indicates that resistance is not determined solely by tumor cells, but emerges through interactions between malignant cells and surrounding stromal and immune components of the tumor microenvironment.

Cyclooxygenase-2 (COX-2) signaling and its associated COXIS transcriptional program have been implicated in inflammation-driven tumor progression. COX-2–derived prostaglandin E (PGE) promotes epithelial–mesenchymal transition (EMT), stress-adaptation programs, and suppression of anti-tumor immunity. In parallel, single-cell studies have identified a distinct subset of cancer-associated fibroblasts (CAFs) marked by LRRC15 expression. These LRRC15 CAFs arise in response to TGF- signaling, remodel extracellular matrix, and directly impair CD8 T-cell effector function, thereby limiting response to checkpoint blockade. Tumor cells may also adopt stress-response states such as the hypoxia-related drug-resistant (HRDS) program, enabling survival under therapy pressure.

However, most prior work has analyzed these processes using bulk transcriptomics or dissociated cells, which remove spatial architecture. The spatial arrangement of tumor cells, CAFs, and immune cells is increasingly recognized as critical for therapy response, yet the spatial relationship connecting COXIS-high tumor regions, LRRC15 CAF localization, EMT/HRDS programs, and ICB non-responder signatures is not well understood.

-Research objective

This study used spatial transcriptomic data to evaluate how tumor-intrinsic COXIS signaling and microenvironmental resistance programs are organized within intact breast tumor tissue.

We hypothesized that within breast tumors, spatially localized COXIS-high tumor nests are adjacent to LRRC15 CAF-rich stroma, indicating spatially organized tumor–stroma ecosystems associated with therapy resistance.

The overall objective was to determine whether COX-2 signaling is spatially associated with stromal remodeling and immune therapy resistance in breast cancer.

METHODS AND MATERIALS

-Study design

This investigation was designed as a computational spatial transcriptomic analysis. Our study compared gene expression signatures and biological pathways in spatially defined tumor and stromal regions within breast cancer tissue sections.

-Data acquisition and preprocessing

Spatial transcriptomic dataset (10x Genomix Visium) from breast tumors were analyzed. Raw count matrices and spatial coordinate information were imported as a spatial experiment.

Quality control steps included:

- removal of low-quality spots
- normalization of gene expression data
- Dimentionality reduction and Leiden clustering based on the normalized top 5% Highly Variable Genes(HVGs)

Spots were annotated into major tissue compartments using pathology annotations and Leiden clustering i.e. tumor regions and cancer-associated fibroblast (CAF)–rich stromal regions.

- Spatial signature scoring and group comparison

To identify dominant transcriptional programs in an unbiased manner, we performed exploratory gene-signature scoring across all spatial spots without prior knowledge of which signatures would be relevant. Publicly available gene sets were systematically screened and scored, and enrichment patterns were evaluated across spatially defined groups. Signatures were not preselected based on expected biology but instead, they were interpreted only after enrichment patterns emerged from the data.

After scoring, CAF-rich and tumor-rich groups were compared statistically, and differences in signature enrichment between the groups were assessed using non-parametric tests.

- Differential expression and pathway enrichment

Differentially expressed genes were identified between tumor-enriched spots vs CAF-enriched spots after which Gene Ontology (GO) and pathway enrichment analysis was used to identify biological processes preferentially activated in each region.
Adjusted p-values were calculated using false discovery rate (FDR) correction. Statistical significance was defined as **adjusted $\mathbf{p < 0.05}$**.

Figures were generated to display pathway enrichment and signature distribution among the Tumor and CAF groups.

DATA LOADING AND EXPLORATION

```r
library(scater)
```

```
Loading required package: SingleCellExperiment

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats


Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

    colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
    colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
    colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
    colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
    colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
    colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
    colWeightedMeans, colWeightedMedians, colWeightedSds,
    colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
    rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
```

```
    rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
    rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
    rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
    rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
    rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
    rowWeightedSds, rowWeightedVars

Loading required package: GenomicRanges

Loading required package: stats4

Loading required package: BiocGenerics

Loading required package: generics


Attaching package: 'generics'

The following objects are masked from 'package:base':

    as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
    setequal, union


Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

    IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

    anyDuplicated, aperm, append, as.data.frame, basename, cbind,
    colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
    get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
    order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
    rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
    unsplit, which.max, which.min

Loading required package: S4Vectors
```

```
Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

    findMatches

The following objects are masked from 'package:base':

    expand.grid, I, unname

Loading required package: IRanges


Attaching package: 'IRanges'

The following object is masked from 'package:grDevices':

    windows

Loading required package: Seqinfo

Loading required package: Biobase

Welcome to Bioconductor

    Vignettes contain introductory material; view with
    'browseVignettes()'. To cite Bioconductor, see
    'citation("Biobase")', and for packages 'citation("pkgname")'.


Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

    rowMedians

The following objects are masked from 'package:matrixStats':

    anyMissing, rowMedians
```

```
Loading required package: scuttle

Loading required package: ggplot2
```

```r
library(scran)
library(scuttle)
library(SpatialExperiment)
library(ggspavis)
library(ComplexHeatmap)
```

```
Loading required package: grid

========================================
ComplexHeatmap version 2.26.0
Bioconductor page: http://bioconductor.org/packages/ComplexHeatmap/
Github page: https://github.com/jokergoo/ComplexHeatmap
Documentation: http://jokergoo.github.io/ComplexHeatmap-reference

If you use it in published research, please cite either one:
- Gu, Z. Complex Heatmap Visualization. iMeta 2022.
- Gu, Z. Complex heatmaps reveal patterns and correlations in multidimensional
    genomic data. Bioinformatics 2016.


The new InteractiveComplexHeatmap package can directly export static
complex heatmaps into an interactive Shiny app with zero effort. Have a try!

This message can be suppressed by:
  suppressPackageStartupMessages(library(ComplexHeatmap))
========================================
```

```r
library(igraph)
```

```


Attaching package: 'igraph'


The following object is masked from 'package:scater':

    normalize
```

```
The following object is masked from 'package:GenomicRanges':

    union

The following object is masked from 'package:IRanges':

    union

The following object is masked from 'package:S4Vectors':

    union

The following objects are masked from 'package:BiocGenerics':

    normalize, path, union

The following objects are masked from 'package:generics':

    components, union

The following objects are masked from 'package:stats':

    decompose, spectrum

The following object is masked from 'package:base':

    union
```

```r
library(patchwork)
library(ggplot2)
library(pals)
```

```r
Filter(function(x) is(get(x), "SpatialExperiment"), ls())
```

```
character(0)
```

```r
data = load(file = "Breast_ST6.RData")
```

```r
speBreast
```

```
class: SpatialExperiment
dim: 17943 355
metadata(0):
assays(1): counts
rownames(17943): ENSG00000187634 ENSG00000188976 ... ENSG00000154620
  ENSG00000012817
rowData names(1): symbol
colnames(355): AAACGAGACGGTTGAT-1 AAAGTGTGATTTATCT-1 ...
  TTGACTATTGTCCGGC-1 TTGCCAAGCAGAACCC-1
colData names(5): in_tissue array_row array_col sample_id annotation
reducedDimNames(0):
mainExpName: NULL
altExpNames(0):
spatialCoords names(2) : pxl_col_in_fullres pxl_row_in_fullres
imgData names(4): sample_id image_id data scaleFactor
```

```r
dim(speBreast)
```

```
[1] 17943   355
```

```r
assayNames(speBreast)
```

```
[1] "counts"
```

```r
head(rowData(speBreast))
```

```
DataFrame with 6 rows and 1 column
                     symbol
                <character>
ENSG00000187634      SAMD11
ENSG00000188976       NOC2L
ENSG00000187961      KLHL17
ENSG00000187583     PLEKHN1
ENSG00000187642       PERM1
ENSG00000188290        HES4
```

```r
head(colData(speBreast))
```

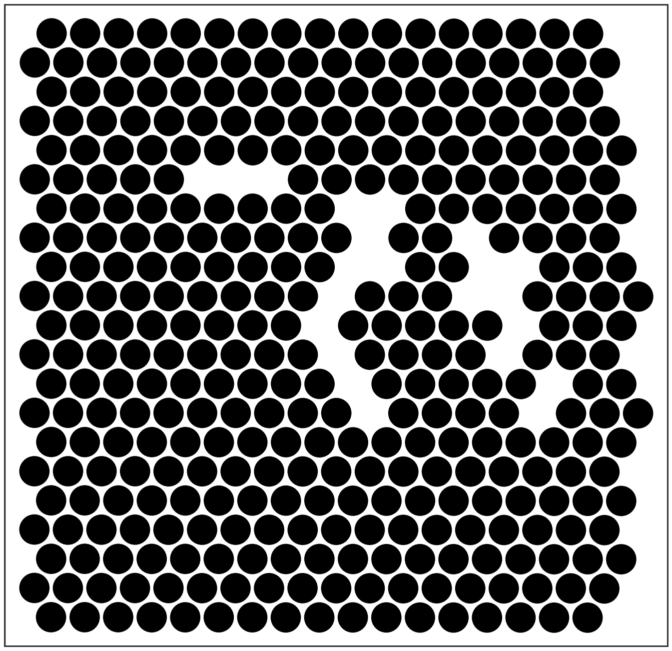
```
DataFrame with 6 rows and 5 columns
                   in_tissue array_row array_col   sample_id  annotation
                   <logical> <integer> <integer> <character> <character>
AAACGAGACGGTTGAT-1      TRUE        35        79    sample01      stroma
AAAGTGTGATTTATCT-1      TRUE        44        94    sample01      stroma
AAAGTTGACTCCCGTA-1      TRUE        42        96    sample01      stroma
AACCCTACTGTCAATA-1      TRUE        40        96    sample01         CAF
AACCGCTAAGGGATGC-1      TRUE        46        92    sample01      stroma
AACTACCCGTTTGTCA-1      TRUE        47       113    sample01    necrosis
```

```r
head(spatialCoords(speBreast))
```

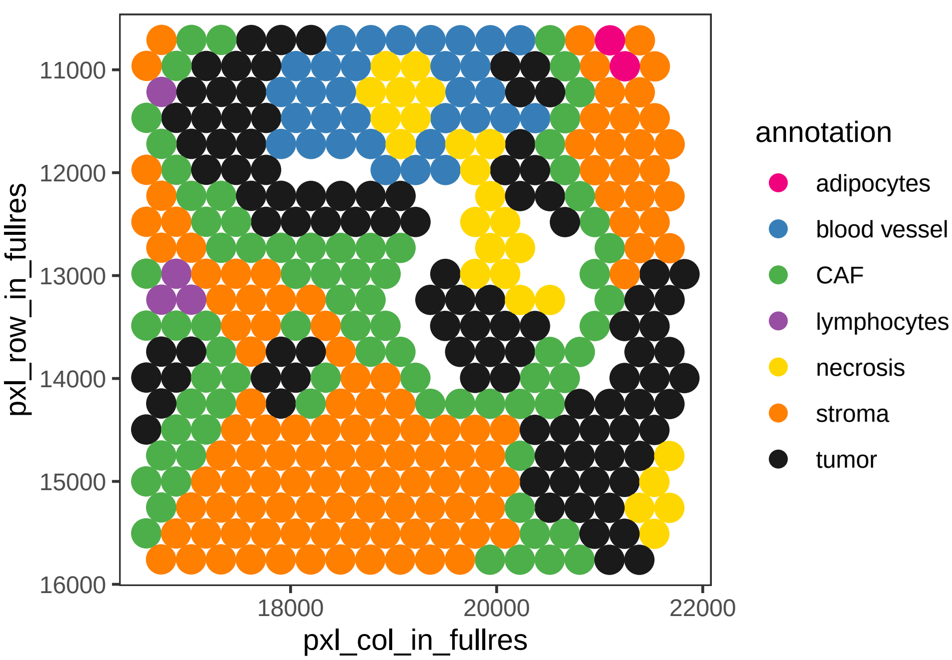
```
                   pxl_col_in_fullres pxl_row_in_fullres
AAACGAGACGGTTGAT-1              16601              12477
AAAGTGTGATTTATCT-1              18775              14750
AAAGTTGACTCCCGTA-1              19066              14246
AACCCTACTGTCAATA-1              19066              13741
AACCGCTAAGGGATGC-1              18485              15255
AACTACCCGTTTGTCA-1              21530              15510
```

DATA VISUALIZATION (ggpavis)

```r
# plot spatial coordinates (spots)
plotCoords(speBreast, point_size = 5)
```



```r
# plot spot by annotation
plotCoords(speBreast, annotate = "annotation",
           point_size = 5, pal = "libd_layer_colors", show_axes = TRUE)
```



Top-left spot annotation

```r
coordinates = spatialCoords(speBreast)
colnames(coordinates) = c("x","y")
coordinates = as.data.frame(coordinates)

head(coordinates)
```

```
                       x     y
AAACGAGACGGTTGAT-1 16601 12477
AAAGTGTGATTTATCT-1 18775 14750
AAAGTTGACTCCCGTA-1 19066 14246
AACCCTACTGTCAATA-1 19066 13741
AACCGCTAAGGGATGC-1 18485 15255
AACTACCCGTTTGTCA-1 21530 15510
```

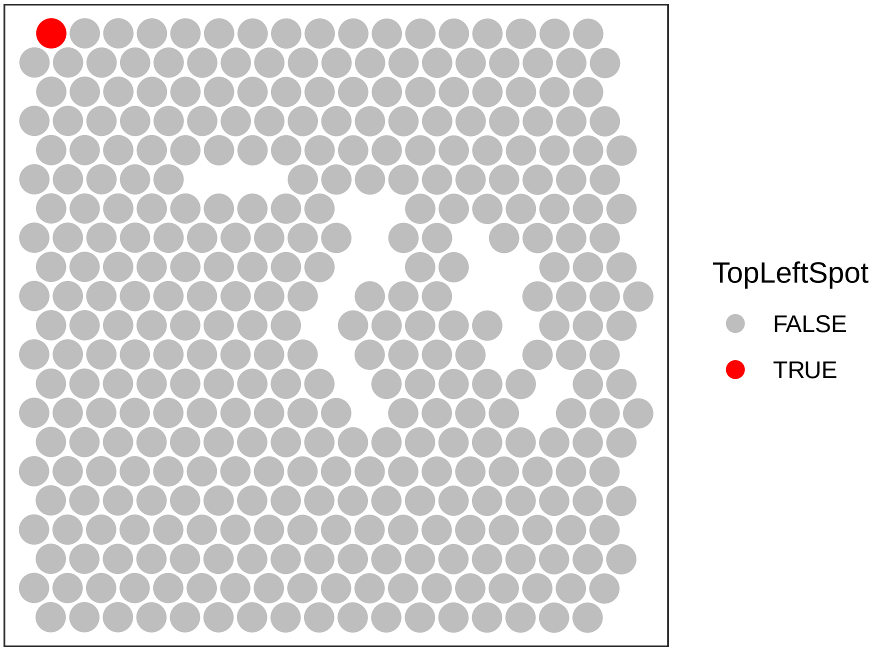
```r
range(coordinates$x)
```

```
[1] 16599 21823
```

```r
range(coordinates$y)
```

```
[1] 10710 15762
```

```r
## top-left spot red
Min_y_Coord <- coordinates$y == min(coordinates$y)
Min_x_Coord <- coordinates$x == min(coordinates$x[Min_y_Coord])

TopLeftSpot <- rownames(coordinates)[which(Min_x_Coord & Min_y_Coord)]
colData(speBreast)[, "TopLeftSpot"] <- as.character(colnames(speBreast)==TopLeftSpot)

plotCoords(speBreast, annotate = "TopLeftSpot", point_size = 5, pal = c("grey", "red"))
```



QUALITY CONTROL (scater library)

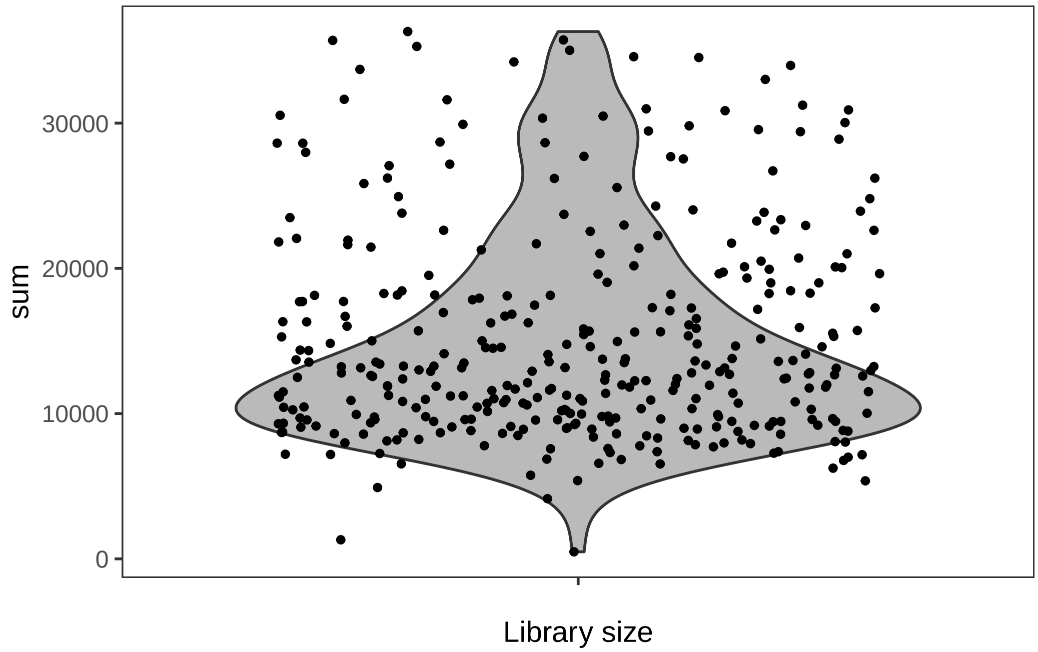
```
speBreast <- addPerCellQC(speBreast)
head(colData(speBreast))
```

```
DataFrame with 6 rows and 9 columns
                     in_tissue array_row array_col   sample_id  annotation
                     <logical> <integer> <integer> <character> <character>
AAACGAGACGGTTGAT-1        TRUE        35        79    sample01      stroma
AAAGTGTGATTTATCT-1        TRUE        44        94    sample01      stroma
AAAGTTGACTCCCGTA-1        TRUE        42        96    sample01      stroma
AACCCTACTGTCAATA-1        TRUE        40        96    sample01         CAF
AACCGCTAAGGGATGC-1        TRUE        46        92    sample01      stroma
AACTACCCGTTTGTCA-1        TRUE        47       113    sample01    necrosis
                     TopLeftSpot       sum  detected       total
                     <character> <numeric> <integer>   <numeric>
AAACGAGACGGTTGAT-1         FALSE     11825      4856       11825
AAAGTGTGATTTATCT-1         FALSE      8121      3599        8121
AAAGTTGACTCCCGTA-1         FALSE      9967      4491        9967
AACCCTACTGTCAATA-1         FALSE     13141      5351       13141
AACCGCTAAGGGATGC-1         FALSE      9794      4079        9794
```

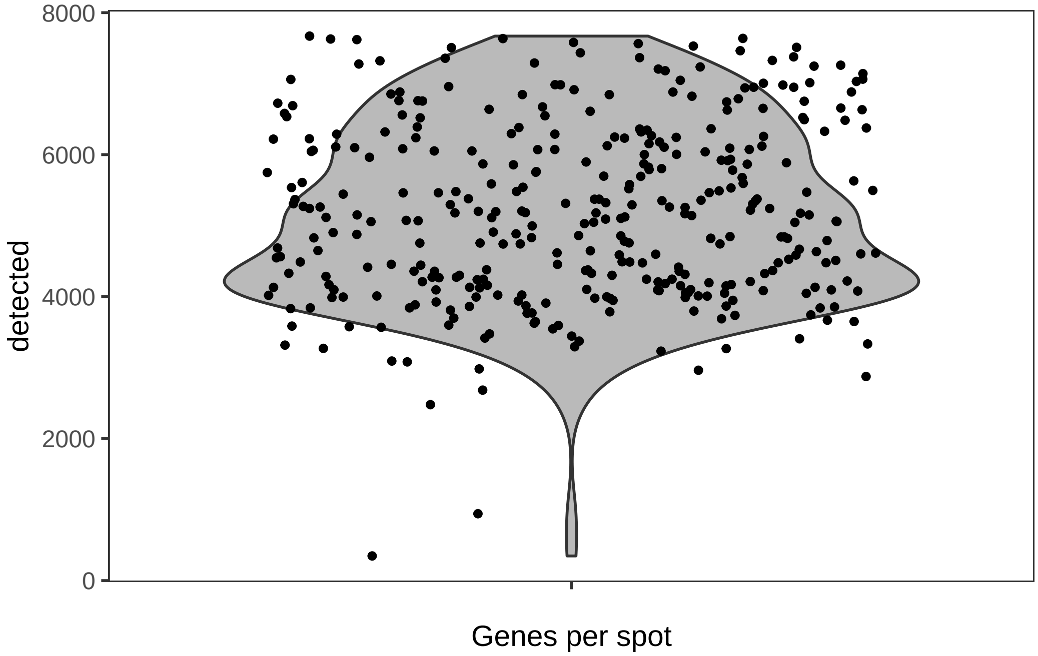
```
AACTACCCGTTTGTCA-1        FALSE      31607       7247      31607
```

QC metrics Visualization

```
plotObsQC(speBreast, plot_type = "violin", x_metric = "sum", point_size = 1) +
  xlab("Library size")
```

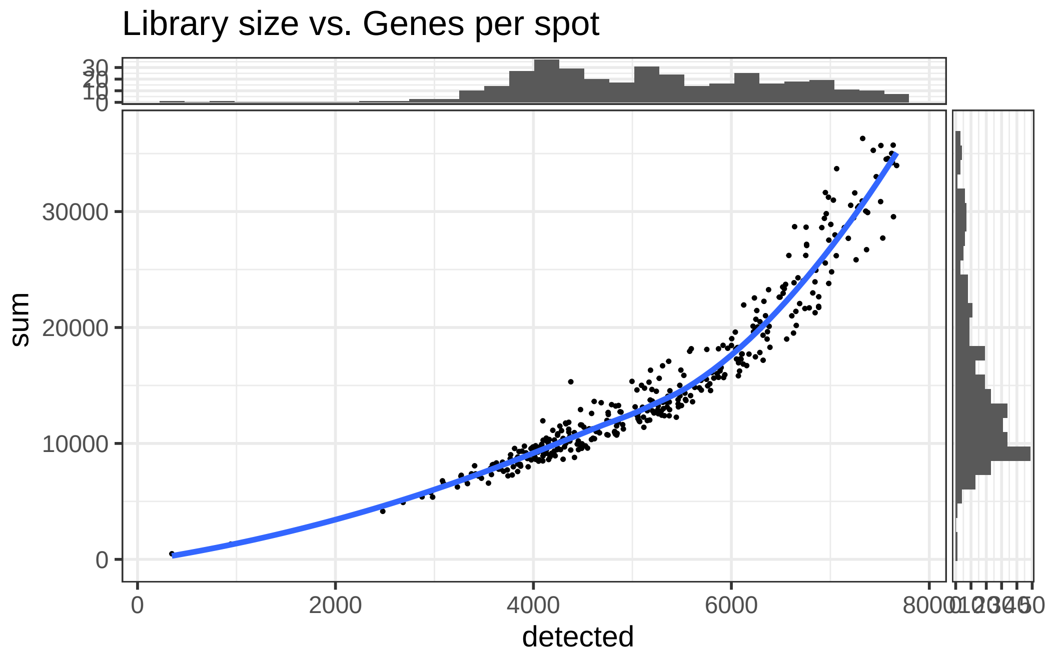



```
plotObsQC(speBreast, plot_type = "violin", x_metric = "detected", point_size = 1) +
  xlab("Genes per spot")
```




```
plotObsQC(
  speBreast, plot_type = "scatter", x_metric = "detected", y_metric = "sum") +
  ggtitle("Library size vs. Genes per spot") + labs(x = "detected", y = "sum")
```

```
`geom_smooth()` using formula = 'y ~ x'
`stat_xsidebin()` using `bins = 30`. Pick better value `binwidth`.
`stat_ysidebin()` using `bins = 30`. Pick better value `binwidth`.
```

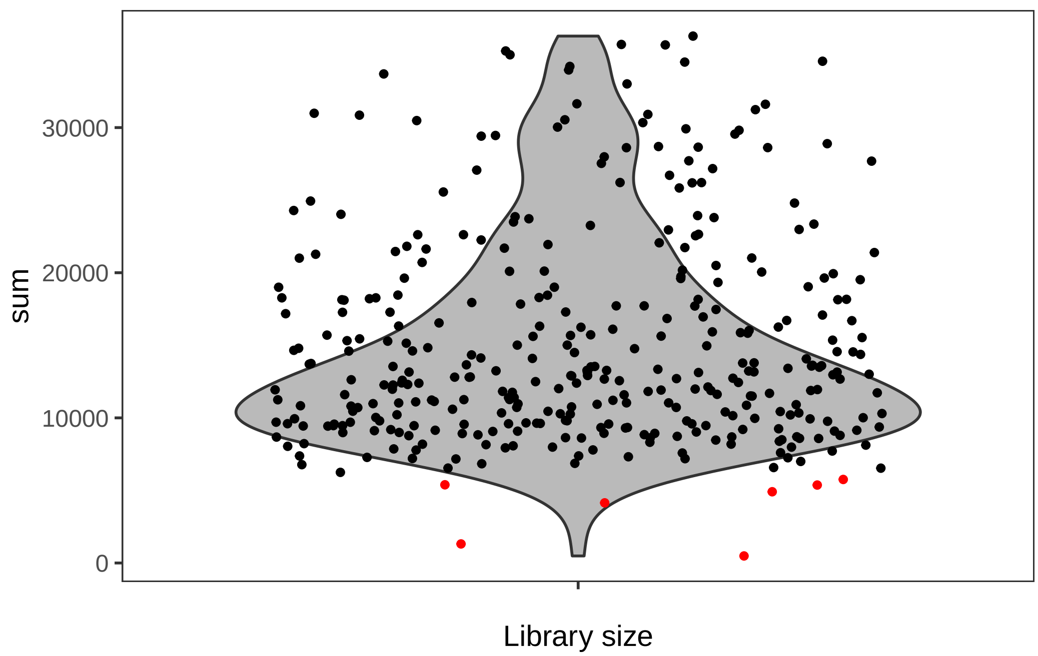


Threshold selection.

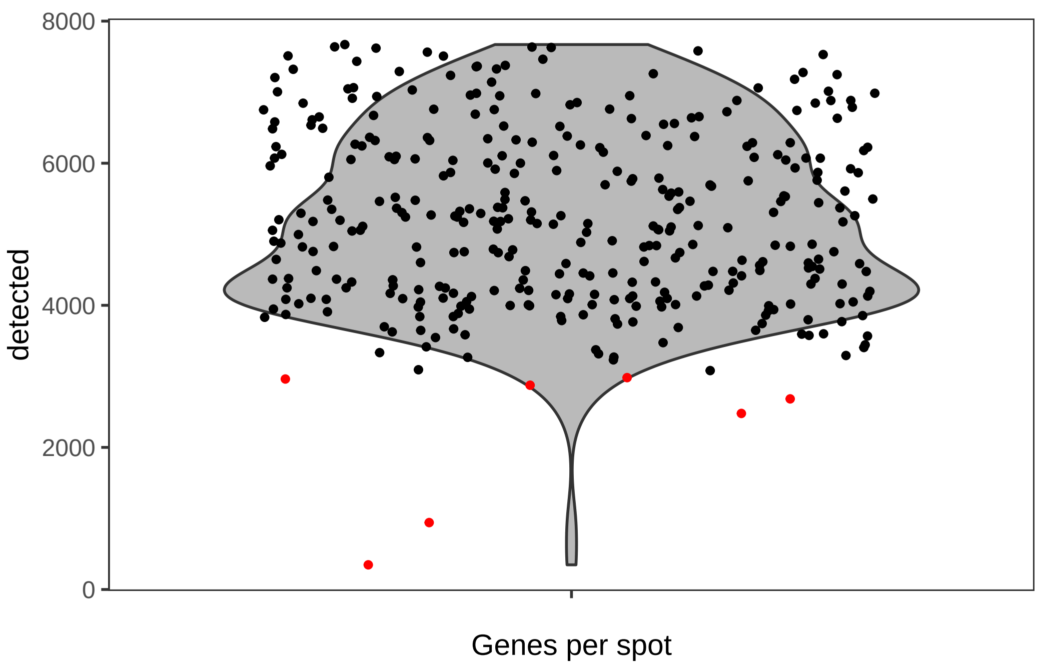
```r
speBreast$qc_lib_size <- speBreast$sum < 6000
speBreast$qc_detected <- speBreast$detected < 3000
table(speBreast$qc_detected, speBreast$qc_lib_size)
```

```
        FALSE TRUE
  FALSE   348    0
  TRUE      0    7
```

```r
plotObsQC(speBreast, plot_type = "violin", x_metric = "sum",
          annotate = "qc_lib_size", point_size = 1) + xlab("Library size")
```

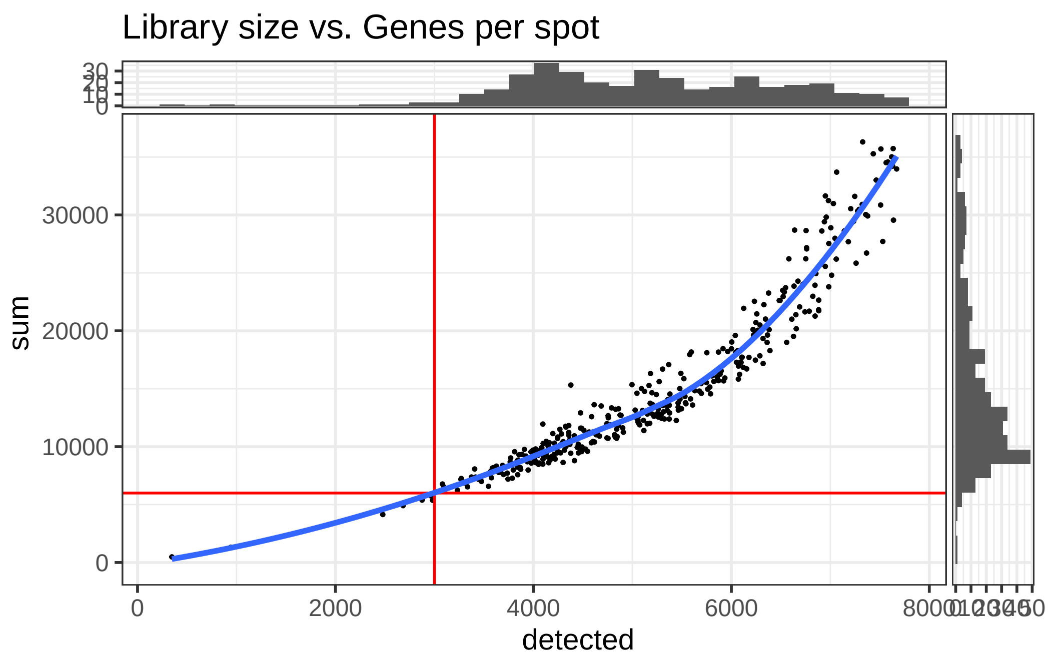



```
plotObsQC(speBreast, plot_type = "violin", x_metric = "detected",
          annotate = "qc_detected", point_size = 1) + xlab("Genes per spot")
```



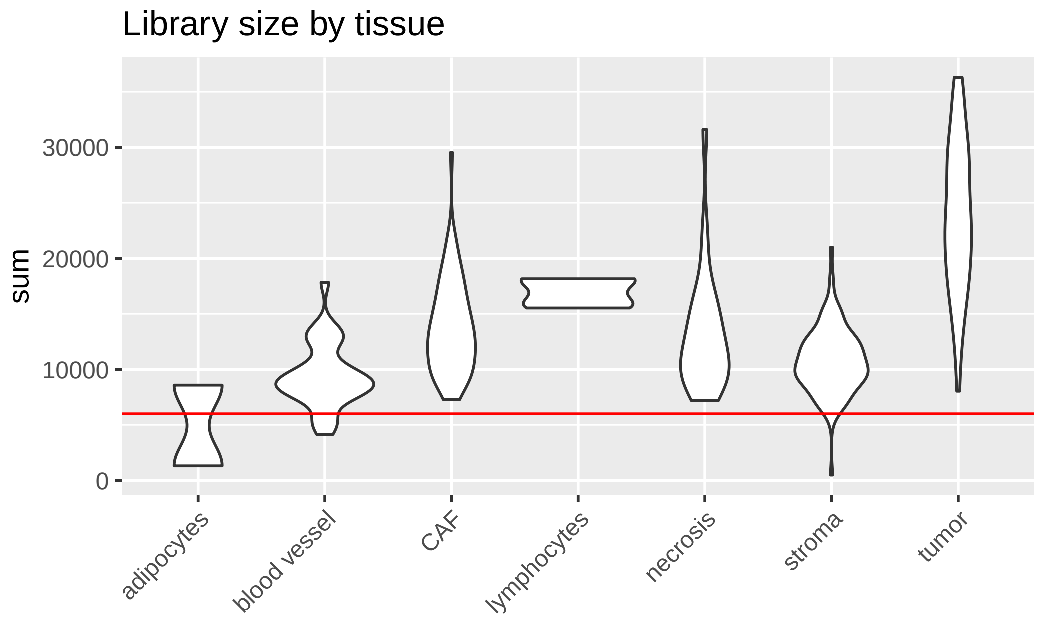
```r
plotObsQC(
  speBreast, plot_type = "scatter", x_metric = "detected",
  y_metric = "sum", x_threshold = 3000, y_threshold = 6000) +
  ggtitle("Library size vs. Genes per spot") + labs(x = "detected", y = "sum")
```

```
`geom_smooth()` using formula = 'y ~ x'
`stat_xsidebin()` using `bins = 30`. Pick better value `binwidth`.
`stat_ysidebin()` using `bins = 30`. Pick better value `binwidth`.
```

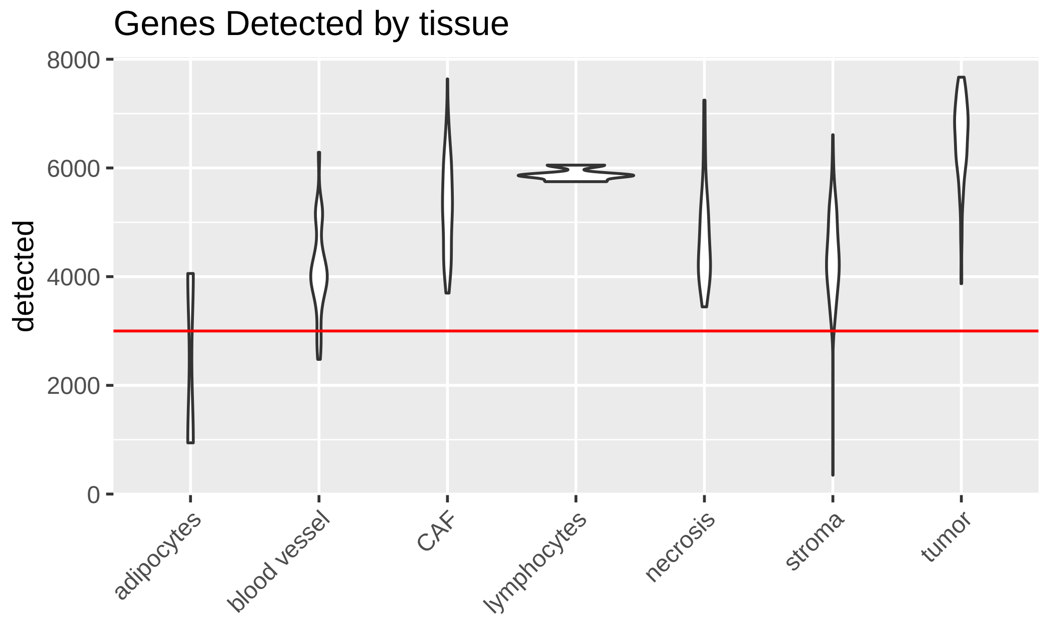


QC metrics visualization by different pathology annotations.

```r
ggplot() + geom_violin(mapping = aes(x = speBreast$annotation, y = speBreast$sum)) +
   theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
   ggtitle("Library size by tissue") + xlab("") + ylab("sum") +
  geom_hline(aes(yintercept = 6000), colour = "red")
```




```
ggplot() + geom_violin(mapping = aes(x = speBreast$annotation, y = speBreast$detected)) +
   theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
   ggtitle("Genes Detected by tissue") + xlab("") + ylab("detected") +
  geom_hline(aes(yintercept = 3000), colour = "red")
```



As observed, Adipocytes have lowest UMIs and genes detected and this could be because they are mostly fat thus containing very little RNA.

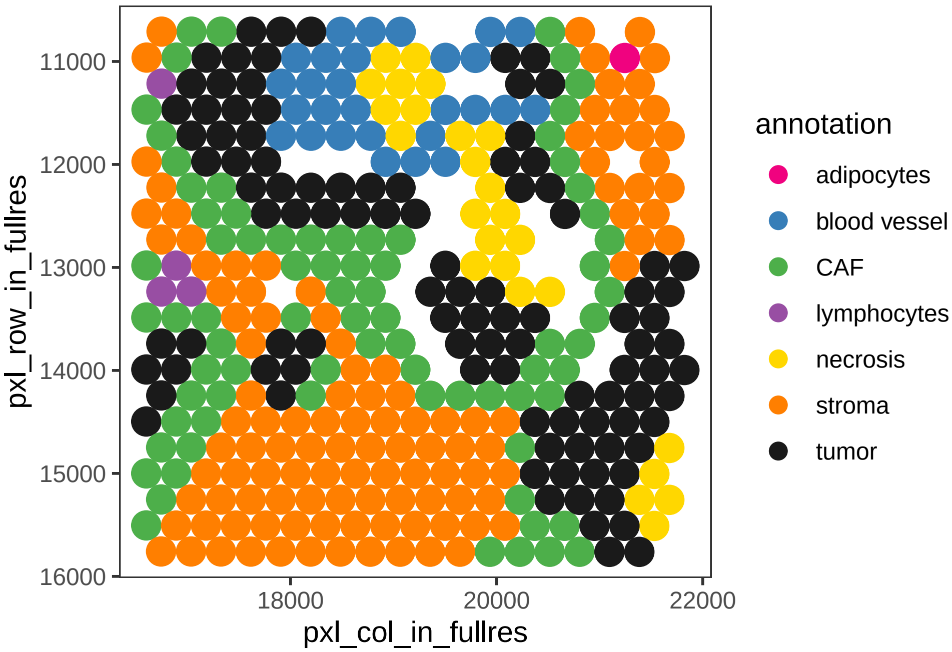
Low-quality spots removal

```
speBreast$outliers <- speBreast$qc_lib_size | speBreast$qc_detected
table(speBreast$outliers, speBreast$annotation)
```

```
        adipocytes blood vessel CAF lymphocytes necrosis stroma tumor
  FALSE          1           28  77           4       25    113   100
  TRUE           1            4   0           0        0      2     0
```

```
speBreast <- speBreast[, !speBreast$outliers]
```

```
plotCoords(speBreast, annotate = "annotation",
           point_size = 5, pal = "libd_layer_colors", show_axes = TRUE)
```



NORMALIZATION (scran)

Getting rid of all the genes whose expression is zero in all the spots.

```
dim(speBreast)
```

```
[1] 17943   348
```

```
speBreast <- speBreast[apply(assay(speBreast) > 0, 1, any),]
```

```
dim(speBreast)
```

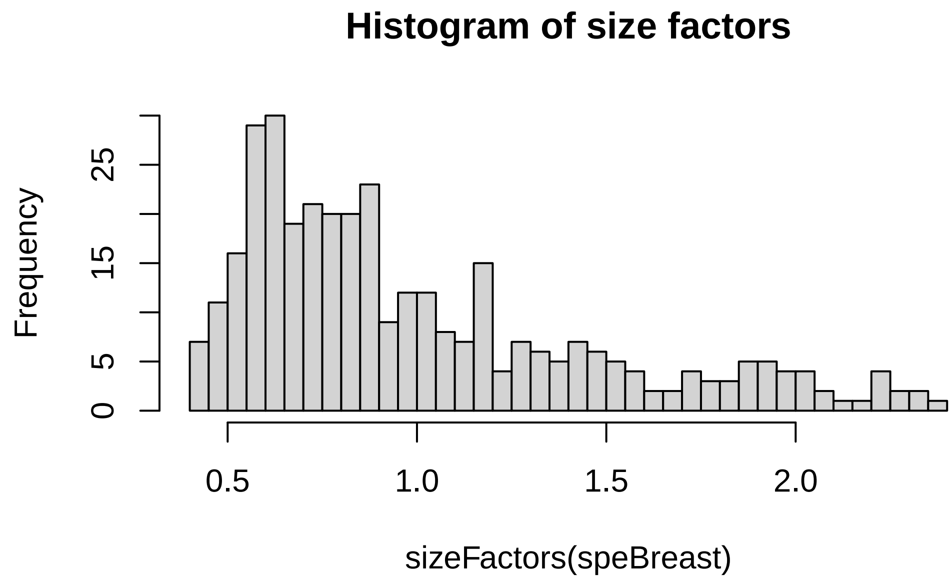
```
[1] 15898   348
```

Then, calculate library size factors and normalize data.

```
# calculate library size factors
speBreast <- computeLibraryFactors(speBreast)
summary(sizeFactors(speBreast))
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
 0.4060  0.6368  0.8539  1.0000  1.2356  2.3608 
```

```
hist(sizeFactors(speBreast), breaks = 50, main = "Histogram of size factors")
```

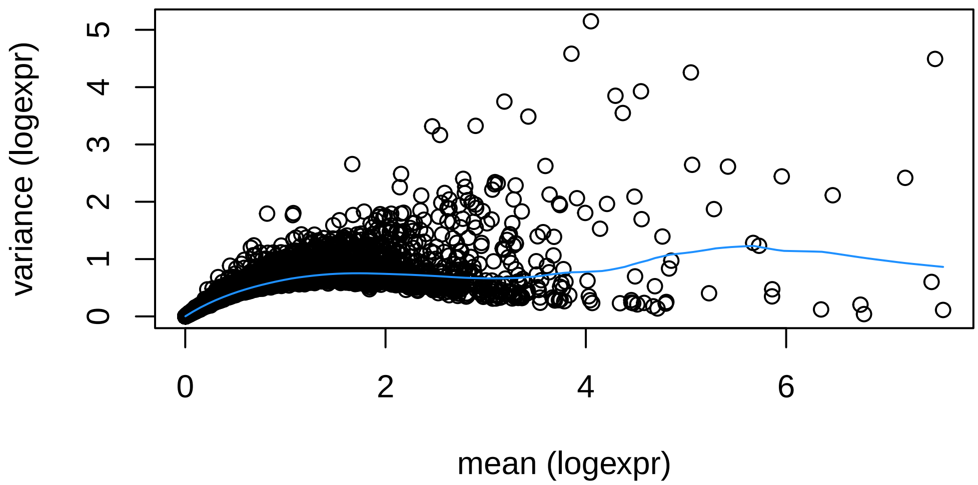



```
# calculate logcounts and store the output in a new assay
speBreast <- logNormCounts(speBreast)
assayNames(speBreast)
```

```
[1] "counts"    "logcounts"
```

Feature selection based on top Highly Variable Genes(HVGs)

```
# fit mean-variance relationship
dec <- modelGeneVar(speBreast)

# visualize mean-variance relationship
plot(dec$mean, dec$total, xlab = "mean (logexpr)", ylab = "variance (logexpr)")
curve(metadata(dec)$trend(x), add = TRUE, col = "dodgerblue")
```




```r
dec
```

```
DataFrame with 15898 rows and 6 columns
                     mean     total      tech         bio   p.value       FDR
                <numeric> <numeric> <numeric>   <numeric> <numeric> <numeric>
ENSG00000187634 0.0552722 0.0624187 0.0603078  0.00211088  0.476748  0.661672
ENSG00000188976 0.5163318 0.3955543 0.4239155 -0.02836120  0.544378  0.661672
ENSG00000187961 0.1152458 0.1198071 0.1209107 -0.00110359  0.506067  0.661672
ENSG00000187583 0.0839241 0.0803361 0.0898814 -0.00954534  0.570223  0.661672
ENSG00000188290 0.7702380 0.6061659 0.5524259  0.05373992  0.435621  0.661672
...                   ...       ...       ...         ...       ...       ...
ENSG00000155961 0.0278028  0.026687 0.0308923 -0.00420525  0.589712  0.661672
ENSG00000155962 0.3101254  0.337756 0.2871967  0.05055956  0.384641  0.661672
ENSG00000185973 0.6094381  0.462645 0.4754351 -0.01278999  0.517875  0.661672
ENSG00000168939 0.1462562  0.148196 0.1503039 -0.00210801  0.509321  0.661672
ENSG00000124333 0.8244133  0.547946 0.5753969 -0.02745135  0.531678  0.661672
```

```r
# select top HVGs
top_hvgs <- getTopHVGs(dec, prop = 0.05)
length(top_hvgs)
```

```
[1] 412
```

Dimensionality reduction

Dimensionality reduction using PCA on the top highly variable genes revealed clear structure in the dataset. In the annotated PCA, spots separated primarily along a tumor–stromal axis on PC1, with additional variation associated with necrotic and vascular regions on PC2. Tumor spots were broadly dispersed, indicating multiple transcriptional subgroups, whereas lymphocytes formed a compact cluster positioned near stromal and CAF spots.

Application of the Leiden algorithm to the PCA neighbor graph identified six transcriptionally coherent clusters that matched the major groupings visible in PCA space. These clusters included distinct tumor-associated groups(4,5), stromal/CAF clusters(1,2,3), and a necrotic/blood vessel cluster(6), with some clusters showing intermediate positions between stromal and CAF populations.

UMAP projections provided a non-linear visualization of the same relationships, showing stromal, CAF, and immune spots forming a connected manifold, while tumor and necrotic/angiogenic clusters formed separate islands. Leiden labeling on the UMAP highlighted the same cluster structure and transitions observed in PCA, confirming consistent grouping of transcriptionally similar spots across both dimensionality reduction approaches.

PCA

```r
set.seed(12345)

speBreast <- runPCA(speBreast, subset_row = top_hvgs)
reducedDimNames(speBreast)
```

```
[1] "PCA"
```

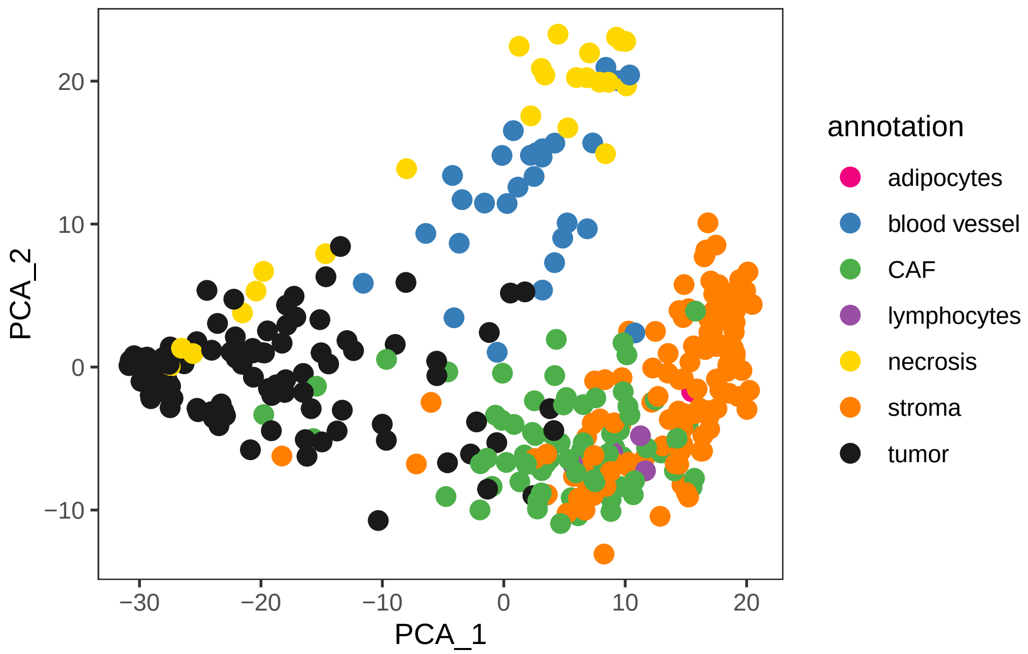
```r
dim(reducedDim(speBreast, "PCA"))
```

```
[1] 348  50
```

```r
colnames(reducedDim(speBreast, "PCA"))
```

```
 [1] "PC1"  "PC2"  "PC3"  "PC4"  "PC5"  "PC6"  "PC7"  "PC8"  "PC9"  "PC10"
[11] "PC11" "PC12" "PC13" "PC14" "PC15" "PC16" "PC17" "PC18" "PC19" "PC20"
[21] "PC21" "PC22" "PC23" "PC24" "PC25" "PC26" "PC27" "PC28" "PC29" "PC30"
[31] "PC31" "PC32" "PC33" "PC34" "PC35" "PC36" "PC37" "PC38" "PC39" "PC40"
[41] "PC41" "PC42" "PC43" "PC44" "PC45" "PC46" "PC47" "PC48" "PC49" "PC50"
```

```r
# plot annotation in PCA reduced dimensions
plotDimRed(speBreast, plot_type = "PCA", annotate = "annotation",
           pal = "libd_layer_colors", point_size = 3)
```



UMAP

```r
speBreast <- runUMAP(speBreast, dimred = "PCA")

reducedDimNames(speBreast)
```

```
[1] "PCA"  "UMAP"
```

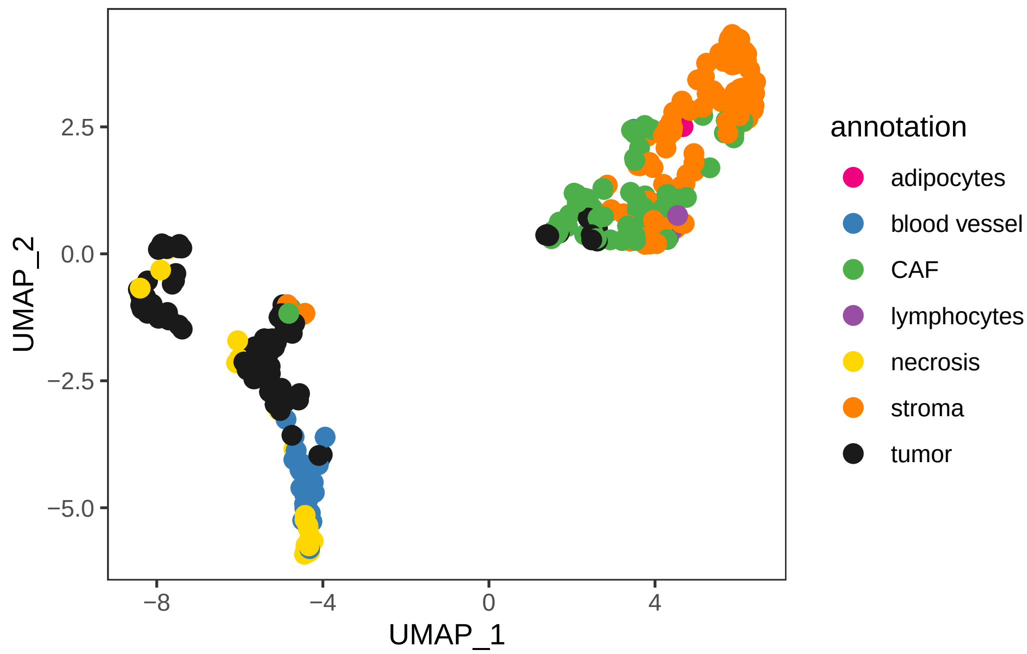
```r
dim(reducedDim(speBreast, "UMAP"))
```

```
[1] 348   2
```

```r
colnames(reducedDim(speBreast, "UMAP"))
```

```
[1] "UMAP1" "UMAP2"
```

```
# plot annotation in UMAP reduced dimensions
plotDimRed(speBreast, plot_type = "UMAP", annotate = "annotation",
           pal = "libd_layer_colors", point_size = 3)
```

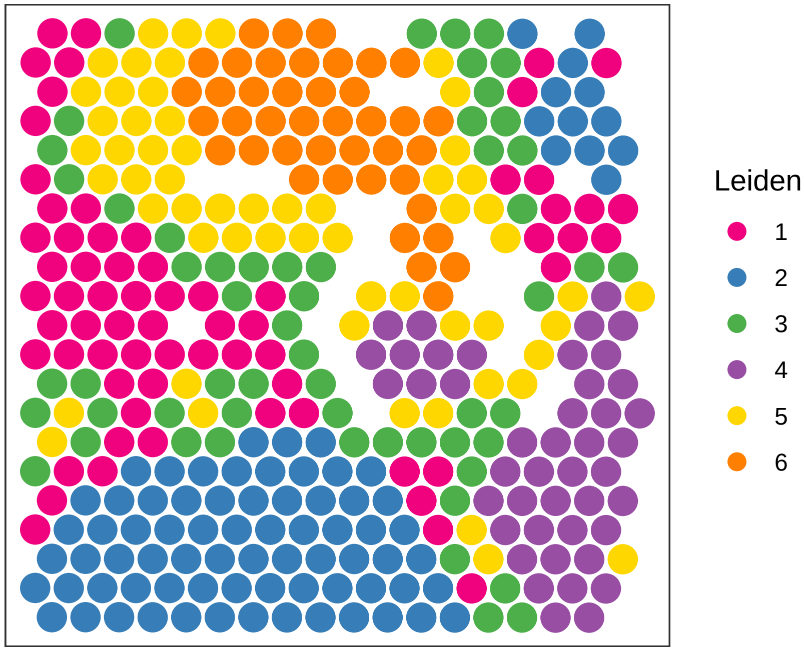


Clustering

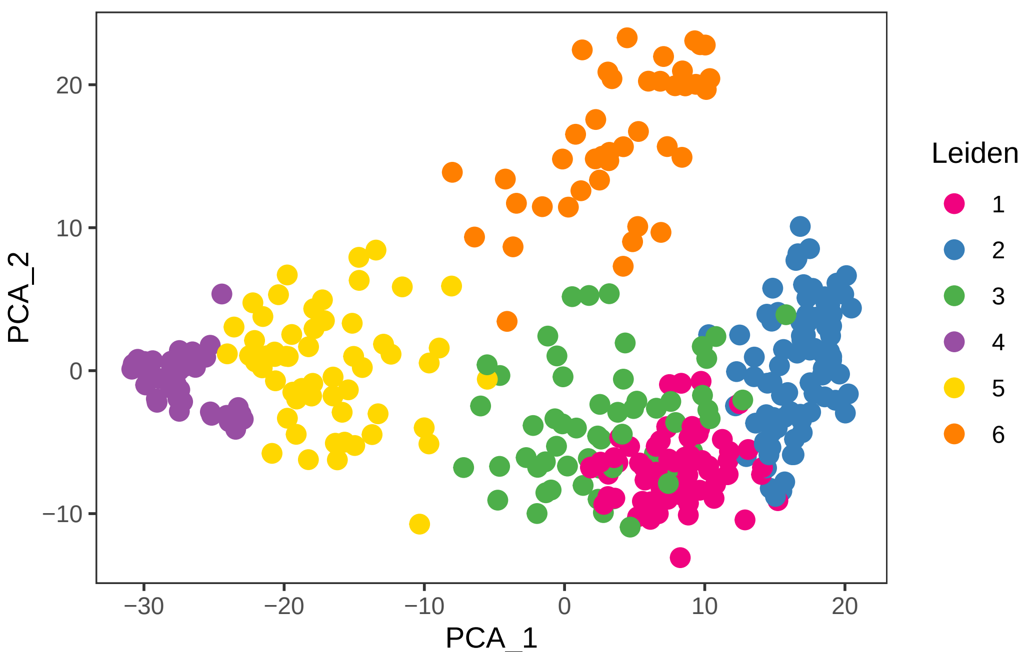
```
k <- 10
g <- buildSNNGraph(speBreast, k = k, use.dimred = "PCA")
k <- igraph::cluster_leiden(g, objective_function="modularity", resolution=1.2)
speBreast$Leiden <- factor(k$membership)
table(speBreast$Leiden)
```

```
 1  2  3  4  5  6
67 82 56 44 58 41
```

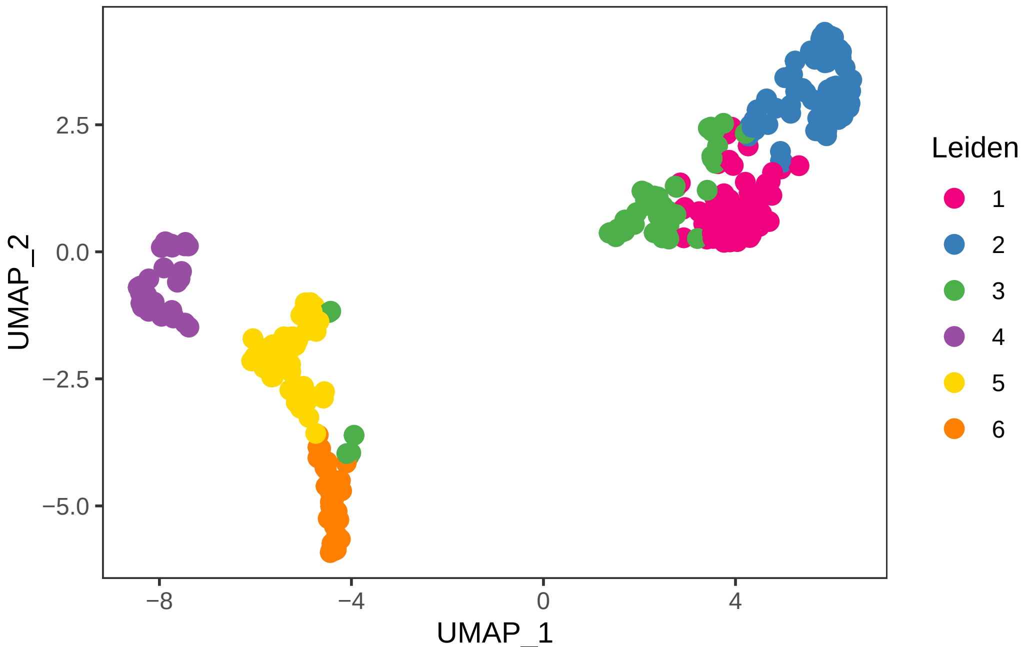
```
# plot clusters in spatial x-y coordinates
plotCoords(speBreast, annotate = "Leiden", pal = "libd_layer_colors", point_size = 5)
```




```r
# plot clusters in PCA reduced dimensions
plotDimRed(speBreast, plot_type = "PCA", annotate = "Leiden",
           pal = "libd_layer_colors", point_size = 3)
```



```r
# plot clusters in UMAP reduced dimensions
plotDimRed(speBreast, plot_type = "UMAP", annotate = "Leiden",
           pal = "libd_layer_colors", point_size = 3)
```




```r
table(
  Leiden = speBreast$Leiden,
  Annotation = speBreast$annotation
)
```

```
      Annotation
Leiden adipocytes blood vessel CAF lymphocytes necrosis stroma tumor
     1          0            0  30           4        0     33     0
     2          1            0   6           0        0     75     0
     3          0            3  37           0        0      4    12
     4          0            0   0           0        4      0    40
     5          0            1   4           0        4      1    48
     6          0           24   0           0       17      0     0
```

Differential Expression

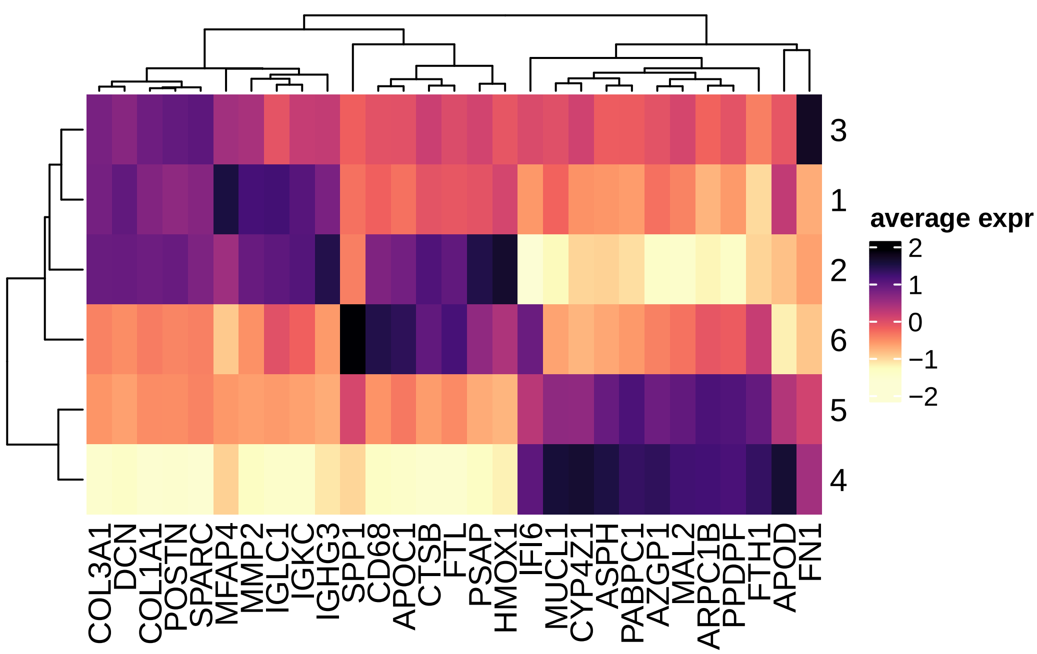
When we perfomed differential expression across Leiden clusters, The heatmap of average gene expression showed distinct expression signatures between groups. Clusters 1 and 3 displayed high expression of extracellular matrix and stromal markers such as **COL1A1, COL3A1, DCN, POSTN, SPARC, and MMP2**, consistent with **CAF-rich regions**. In contrast, clusters 4 and 5 showed higher expression of tumor-associated genes including **MUCL1, MAL2, AZGP1, ASPH, and CYP4Z1**, consistent with **tumor-rich regions**. Immune-related genes such as **IGHG3, IGKC, and CD68** showed localized enrichment in specific clusters, indicating immune infiltration in restricted areas.

```r
mgs <- findMarkers(speBreast, groups = speBreast$Leiden, direction = "up")
top <- lapply(mgs, \(df) rownames(df)[df$Top <= 2])
top <- unique(unlist(top))
length(top)
```

```
[1] 29
```

```r
# compute cluster-wise averages
pbs <- aggregateAcrossCells(speBreast, ids = speBreast$Leiden, subset.row = top,
                            use.assay.type = "logcounts", statistics = "mean")
# use gene symbols as feature names
mtx <- t(assay(pbs))
colnames(mtx) <- rowData(pbs)$symbol
```

```r
library(ComplexHeatmap)
Heatmap(mat = scale(mtx), name = "average expr", col = c("#FCFDD4", rev(viridis::magma(10))))
```



These transcriptional profiles align with the dimensionality reduction findings. Clusters enriched in collagen and matrix genes correspond to the CAF/stromal regions seen in PCA/UMAP space, while immunoglobulin-rich clusters match the lymphocyte and immune populations that localized adjacent to stroma rather than tumor, supporting an immune-excluded architecture. The hypoxia/stress-associated clusters are consistent with the necrotic/angiogenic regions identified along PC2. Overall, the heatmap confirms the presence of distinct tumor, CAF/stromal, immune, and hypoxic/angiogenic compartments previously identified by PCA, UMAP, and Leiden clustering.

RESULTS AND DISCUSSION

We proposed that breast tumors contain spatially localized COXIS-high tumor nests adjacent to LRRC15 CAF-rich stroma, forming organized tumor–stroma ecosystems potentially linked to therapy resistance.
Dimensionality reduction and clustering revealed distinct transcriptional regions, which we broadly grouped as CAF-rich (clusters 1 and 3) and tumor-rich (clusters 4 and 5) based on marker-expression patterns in the heatmap.

To compare these compartments, we performed gene-signature scoring for tumor and CAF groups we obtained after clustering.
We also carried out Gene Ontology pathway enrichment analysis separately in CAF-rich and tumor-rich groups to identify dominant biological processes. These analyses allowed us to assess how tumor and CAF regions differ transcriptionally and functionally, and whether their spatial organization supported our hypothesis.

```
# 1) Defining CAF vs tumor groups

caf_clusters <- c("1","3")
tumor_clusters <- c("4","5")

group <- ifelse(speBreast$Leiden %in% caf_clusters,
                "CAF",
                ifelse(speBreast$Leiden %in% tumor_clusters,
                       "Tumor",
                       NA))

speBreast$group <- group

# keeping only CAF and tumor
spe_sub_sig <- speBreast[, !is.na(speBreast$group)]

spe_sub_GO <- speBreast[, !is.na(speBreast$group)]
```

GENE-SIGNATURE SCORING

```r
library(signifinder)
```

```r
library(org.Hs.eg.db)
```

```
Loading required package: AnnotationDbi
```

```r
symbols <- mapIds(
  org.Hs.eg.db,
  keys = rownames(spe_sub_sig),
  keytype = "ENSEMBL",
  column = "SYMBOL",
  multiVals = "first"
)
```

```
'select()' returned 1:many mapping between keys and columns
```

```r
rownames(spe_sub_sig) <- symbols
```

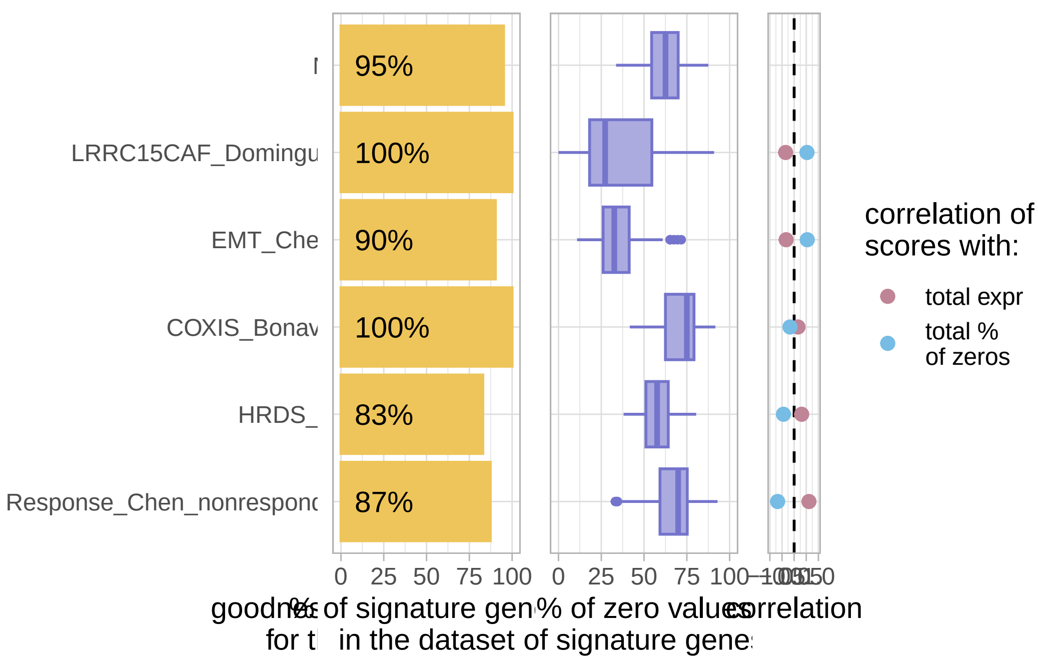
```r
spe_sig <- multipleSign(
  dataset = spe_sub_sig,
  inputType = "rnaseq",
  tissue = "breast",
  whichAssay = "logcounts"   # <<< replace with the normalized assay you actually have
)
```

```
EMTSignCheng is using 90% of signature genes

HRDSSign is using 83% of signature genes

LRRC15CAFSign is using 100% of signature genes

ICBResponseSign is using 95% of responder signature genes
```

```
ICBResponseSign is using 87% of nonresponder signature genes

Warning in cor(x, sup_resp, method = "pearson"): the standard deviation is zero
Warning in cor(x, sup_resp, method = "pearson"): the standard deviation is zero
Warning in cor(x, sup_resp, method = "pearson"): the standard deviation is zero

COXISSign is using 100% of signature genes
```

```r
#checking signature goodness
evaluationSignPlot(spe_sig, whichAssay = "logcounts")
```

```
Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_bar()`).

Warning: Removed 1 row containing missing values or values outside the scale range
(`geom_text()`).

Warning: Removed 1 row containing missing values or values outside the scale range
(`geom_point()`).
Removed 1 row containing missing values or values outside the scale range
(`geom_point()`).
```

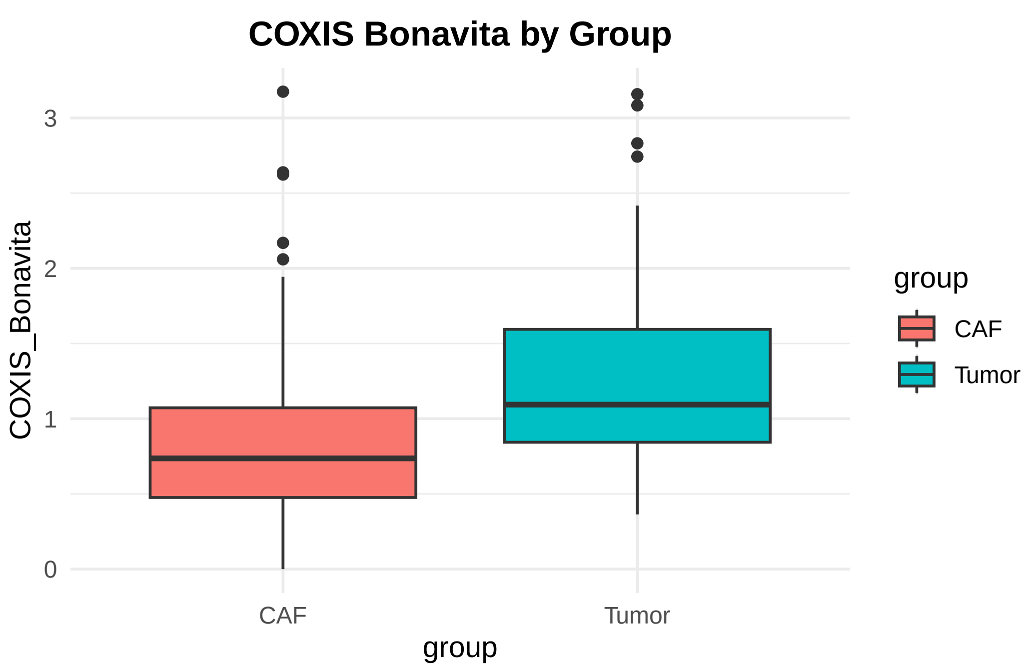


Several signatures showed high goodness-of-fit in the dataset, with LRRC15-CAF, EMT, and COXIS scoring highest and most signatures having >80% gene coverage. Zero-value proportions varied across signatures, and correlations with total expression or sparsity were generally low, indicating minimal technical bias.

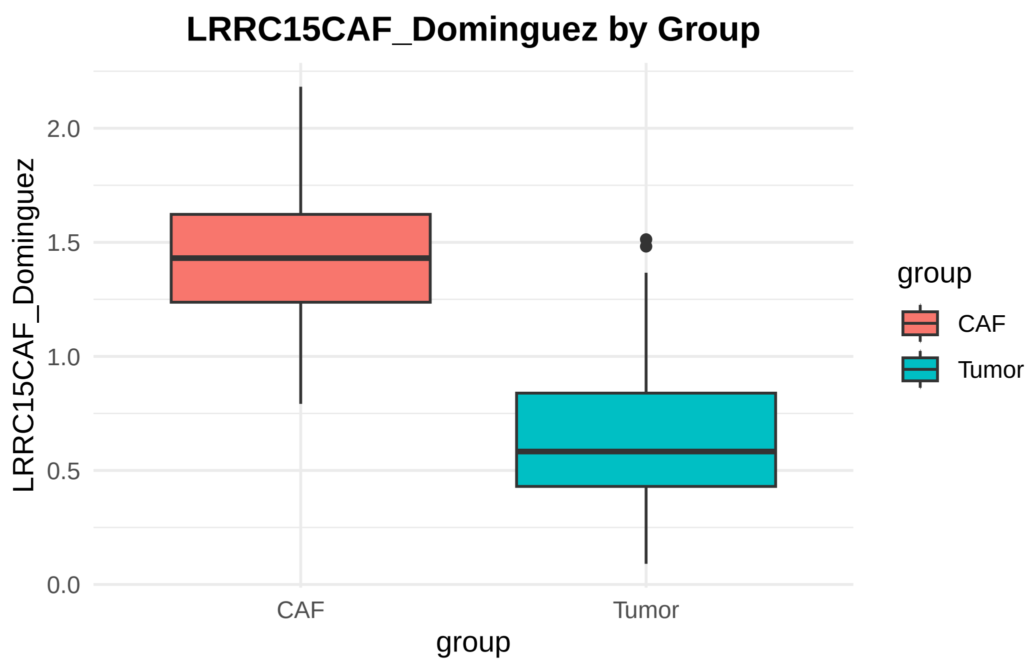
We then identified which biological programs were preferentially enriched in each compartment by comparing the two groups based on the gene-signatures scored.

Gene-signature scoring showed clear differences between the two groups. COXIS scores were significantly higher in tumor-rich clusters (4 and 5), while LRRC15-CAF and EMT signatures were significantly higher in CAF-rich clusters (1 and 3). Thus, tumor regions preferentially scored high for COXIS, whereas CAF-rich regions preferentially scored high for LRRC15-CAF and EMT programs.

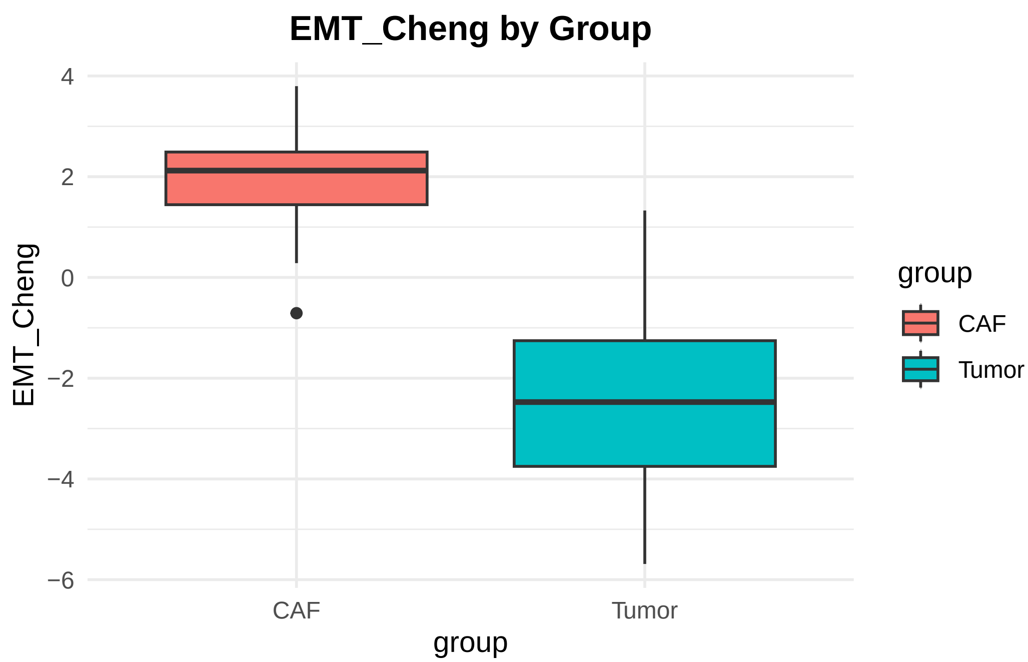
```
df <- as.data.frame(colData(spe_sig))

ggplot(df, aes(x = group, y =
COXIS_Bonavita, fill = group)) +
    geom_boxplot() +
    theme_minimal() +
  labs(title = "COXIS Bonavita by Group") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
```



```
ggplot(df, aes(x = group, y = LRRC15CAF_Dominguez, fill = group)) +
    geom_boxplot() +
    theme_minimal() +
  labs(title = "LRRC15CAF_Dominguez by Group") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
```




```
ggplot(df, aes(x = group, y = EMT_Cheng, fill = group)) +
    geom_boxplot() +
    theme_minimal() +
  labs(title = "EMT_Cheng by Group") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
```




```
wilcox.test(COXIS_Bonavita ~ group, data = df)
```

```
    Wilcoxon rank sum test with continuity correction

data:  COXIS_Bonavita by group
W = 3310.5, p-value = 1.104e-09
alternative hypothesis: true location shift is not equal to 0
```

```
wilcox.test(LRRC15CAF_Dominguez ~ group, data = df)
```

```
    Wilcoxon rank sum test with continuity correction

data:  LRRC15CAF_Dominguez by group
W = 12101, p-value < 2.2e-16
alternative hypothesis: true location shift is not equal to 0
```

```
wilcox.test(EMT_Cheng ~ group, data = df)
```

```
	Wilcoxon rank sum test with continuity correction

data:  EMT_Cheng by group
W = 12454, p-value < 2.2e-16
alternative hypothesis: true location shift is not equal to 0
```

The divergent signature profiles indicate distinct but coordinated tumor and stromal programs. Elevated COXIS scores in tumor-rich clusters are consistent with prior work showing activation of COXIS-related inflammatory and oxidative stress programs in tumor nests associated with therapeutic adaptation (***Bonavita et al., Cell 2020***). Higher LRRC15-CAF signatures in CAF-rich clusters agree with studies identifying LRRC15 CAFs as a specialized fibroblast population enriched in treated and resistant tumors (***Dominguez et al., Cancer Cell 2020; Elyada et al., Cancer Cell 2019***). Concurrent enrichment of EMT signatures within CAF-rich regions is consistent with the established role of CAFs in promoting EMT and invasion (***Nieto et al., Cell 2016; Cheng et al., Nat Commun 2021***). Taken together, these findings support a model in which COXIS-high tumor nests coexist with LRRC15 CAF-rich stroma as spatially organized tumor–stroma ecosystems linked to stress adaptation and therapy resistance (***Kieffer et al., Cancer Discovery 2020***).

GENE ONTOLOGY PATHWAY ENRICHMENT

```r
library(clusterProfiler)
```

```
clusterProfiler v4.18.4 Learn more at https://yulab-smu.top/contribution-knowledge-mining/

Please cite:

S Xu, E Hu, Y Cai, Z Xie, X Luo, L Zhan, W Tang, Q Wang, B Liu, R Wang,
W Xie, T Wu, L Xie, G Yu. Using clusterProfiler to characterize
multiomics data. Nature Protocols. 2024, 19(11):3292-3320


Attaching package: 'clusterProfiler'
```

```
The following object is masked from 'package:AnnotationDbi':

    select


The following object is masked from 'package:igraph':

    simplify


The following object is masked from 'package:IRanges':

    slice


The following object is masked from 'package:S4Vectors':

    rename


The following object is masked from 'package:stats':

    filter
```

```
# Differential expression

de <- findMarkers(
  spe_sub_GO,
  groups = spe_sub_GO$group,
  direction = "up"   # both up and down
)

# Inspection

caf_vs_tumor <- de$CAF[ de$CAF$FDR < 0.05, ]   # genes up in CAF relative to tumor
tumor_vs_caf <- de$Tumor[ de$Tumor$FDR < 0.05, ] # genes up in tumor relative to CAF

head(caf_vs_tumor)
```

```
DataFrame with 6 rows and 5 columns
                      Top     p.value         FDR summary.logFC logFC.Tumor
                <integer>   <numeric>   <numeric>     <numeric>   <numeric>
ENSG00000132386         1 6.25708e-64 9.94750e-60       2.20036     2.20036
ENSG00000163520         2 1.31411e-62 1.04458e-58       2.21299     2.21299
ENSG00000011465         3 2.00439e-62 1.06219e-58       2.99505     2.99505
```

```
ENSG00000159403         4 2.84860e-61 1.13218e-57       2.43602     2.43602
ENSG00000087245         5 3.00082e-60 9.54140e-57       2.53610     2.53610
ENSG00000038427         6 1.35812e-57 3.59857e-54       2.13361     2.13361
```

```r
head(tumor_vs_caf)
```

```
DataFrame with 6 rows and 5 columns
                      Top     p.value         FDR summary.logFC logFC.CAF
                <integer>   <numeric>   <numeric>     <numeric> <numeric>
ENSG00000070756         1 2.16872e-69 3.44784e-65       1.07729   1.07729
ENSG00000167996         2 1.50743e-67 1.19826e-63       1.41395   1.41395
ENSG00000125534         3 1.01889e-60 5.39946e-57       1.36827   1.36827
ENSG00000160862         4 1.27510e-59 5.06788e-56       1.70512   1.70512
ENSG00000130429         5 4.29151e-59 1.36453e-55       1.10969   1.10969
ENSG00000272398         6 8.59861e-56 2.27835e-52       1.53675   1.53675
```

```r
# Ensemble to ENTREIDz

caf_genes <- rownames(caf_vs_tumor)

caf_entrez <- bitr(
  caf_genes,
  fromType = "ENSEMBL",
  toType = "ENTREZID",
  OrgDb = org.Hs.eg.db
)
```

```
'select()' returned 1:many mapping between keys and columns

Warning in bitr(caf_genes, fromType = "ENSEMBL", toType = "ENTREZID", OrgDb =
org.Hs.eg.db): 0.32% of input gene IDs are fail to map...
```

```r
tumor_genes <- rownames(tumor_vs_caf)

tumor_entrez <- bitr(
  tumor_genes,
  fromType = "ENSEMBL",
  toType = "ENTREZID",
  OrgDb = org.Hs.eg.db
)
```

```
'select()' returned 1:many mapping between keys and columns

Warning in bitr(tumor_genes, fromType = "ENSEMBL", toType = "ENTREZID", : 0.34%
of input gene IDs are fail to map...
```

```r
# Enrichment -----------

# CAF

ego_caf <- enrichGO(
  gene = caf_entrez$ENTREZID,
  OrgDb = org.Hs.eg.db,
  ont = "BP",
  pAdjustMethod = "BH",
  readable = TRUE
)

# TUMOR

ego_tumor <- enrichGO(
  gene = tumor_entrez$ENTREZID,
  OrgDb = org.Hs.eg.db,
  ont = "BP",
  readable = TRUE
)

head(ego_caf)
```

```
                   ID                            Description GeneRatio
GO:0007159 GO:0007159            leukocyte cell-cell adhesion  165/2321
GO:0050867 GO:0050867 positive regulation of cell activation  153/2321
GO:0070661 GO:0070661                 leukocyte proliferation  142/2321
GO:0006935 GO:0006935                              chemotaxis  168/2321
GO:0030198 GO:0030198       extracellular matrix organization  135/2321
GO:0043062 GO:0043062    extracellular structure organization  135/2321
             BgRatio RichFactor FoldEnrichment   zScore       pvalue
GO:0007159 439/18860  0.3758542       3.054119 16.31337 6.451480e-43
GO:0050867 401/18860  0.3815461       3.100371 15.92599 9.313542e-41
GO:0070661 356/18860  0.3988764       3.241193 15.99240 1.815114e-40
GO:0006935 474/18860  0.3544304       2.880033 15.52935 8.100015e-40
GO:0030198 331/18860  0.4078550       3.314151 15.91190 8.963155e-40
GO:0043062 332/18860  0.4066265       3.304169 15.86760 1.333967e-39
```

```
               p.adjust       qvalue
GO:0007159 3.883146e-39 2.602323e-39
GO:0050867 2.802911e-37 1.878394e-37
GO:0070661 3.641724e-37 2.440532e-37
GO:0006935 1.078985e-36 7.230907e-37
GO:0030198 1.078985e-36 7.230907e-37
GO:0043062 1.300925e-36 8.718259e-37

GO:0007159               HLA-DRA/HLA-DPA1/CD74/HLA-DPB1/THY1/CXCL12/CORO1A/PTPRC/LAPTM5/TGFB1
E/FGL2/HLA-DMA/HLA-DMB/GPNMB/JAK3/IL7R/CCL19/CCL5/CD4/IGF1/IL6ST/ITGB2/TGFBR2/CERCAM/CAV1/ITG
DOB/EGR3/CHST2/DPP4/SLAMF1/CBLB/NR5A2/PTPN6/SKAP1/NLRP3/PDCD1LG2/ITGAL/HLA-
DOA/KLRK1/KLRC4-KLRK1/HAS2/CARD11/SPN/CD80/HAVCR2/ICOS/CD5/CD177/CBFB/CD1D/SIRPG/CD300A/TSPAN
GO:0050867
DRA/HLA-DPA1/CD74/HLA-DPB1/THY1/CORO1A/THBS1/MMP14/PTPRC/GAS6/ACTA2/TGFB1/HLA-
E/SPI1/TYROBP/HLA-DMA/HLA-DMB/AXL/IL7R/CCL19/CCL5/CD4/IGF1/IL6ST/SH3KBP1/TGFBR2/FCGR3A/CD40/C
DOB/EGR3/DPP4/IL33/SLAMF1/NR5A2/PTPN6/NLRP3/PDCD1LG2/HLA-DOA/CD226/KLRK1/KLRC4-
KLRK1/CARD11/SPN/CD80/HAVCR2/ICOS/CD5/CBFB/CD1D/SIRPG/BCL6/TNFSF13B/LILRB1/TOX/SELENOK/GLI2/T
GO:0070661
DPA1/CD74/HLA-DPB1/CNN2/CORO1A/PTPRC/GSTP1/LAPTM5/TGFB1/LST1/CD37/HLA-E/TYROBP/HLA-
DMB/GPNMB/CSF1R/IL7R/MZB1/GREM1/CCL19/CCL5/CD4/IGF1/IL6ST/TGFBR2/FCGR3A/CSF2RB/CD40/NCKAP1L/
GO:0006935 MMP2/C3/NBL1/MICOS10-NBL1/CD74/DUSP1/CCN1/PDGFRB/CXCL12/CORO1A/THBS1/LSP1/GAS6/GST
KLRK1/SPN/ADGRE2/ROBO3/PREX1/NEDD9/CCL4L1/ANGPT2/IL6R/MPP1/CXCR4/RHOG/FGFR1/PTPRO/FGF1/SEMA3A
GO:0030198
GO:0043062
           Count
GO:0007159   165
GO:0050867   153
GO:0070661   142
GO:0006935   168
GO:0030198   135
GO:0043062   135
```

```
head(ego_tumor)
```

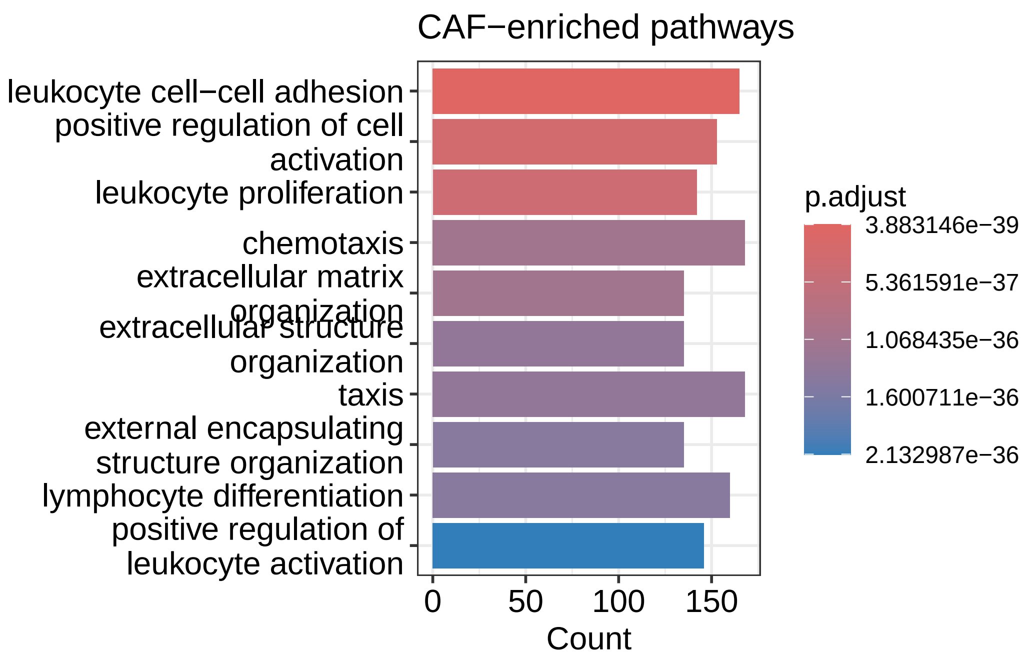
```
                   ID                                        Description
GO:1901293 GO:1901293          nucleoside phosphate biosynthetic process
GO:0031647 GO:0031647                    regulation of protein stability
GO:0048193 GO:0048193                            Golgi vesicle transport
GO:0009060 GO:0009060                                aerobic respiration
GO:0015980 GO:0015980 energy derivation by oxidation of organic compounds
GO:0006091 GO:0006091      generation of precursor metabolites and energy
           GeneRatio   BgRatio RichFactor FoldEnrichment   zScore       pvalue
```

```
GO:1901293  105/3267 332/18860  0.3162651       1.825760 6.948289 1.043014e-
10
GO:0031647  121/3267 404/18860  0.2995050       1.729006 6.779883 2.014194e-
10
GO:0048193   99/3267 319/18860  0.3103448       1.791584 6.526714 1.112679e-
09
GO:0009060   93/3267 297/18860  0.3131313       1.807670 6.421802 2.129669e-
09
GO:0015980  123/3267 434/18860  0.2834101       1.636093 6.136474 5.978871e-
09
GO:0006091  136/3267 494/18860  0.2753036       1.589295 6.075164 7.168460e-
09
               p.adjust       qvalue
GO:1901293 6.214797e-07 5.263408e-07
GO:0031647 6.214797e-07 5.263408e-07
GO:0048193 2.288781e-06 1.938404e-06
GO:0009060 3.285547e-06 2.782581e-06
GO:0015980 7.372761e-06 6.244106e-06
GO:0006091 7.372761e-06 6.244106e-06

GO:1901293
GO:0031647
5/GTPBP4/COG7/PPP1CA/MFSD8/PPARGC1A/DVL1/USP46/USP7/SUGT1/MAPKAP1/AHSA1/TADA3/CREB1/TSPAN1/S
GO:0048193
GO:0009060
GO:0015980                                                                              ATP5
GO:0006091 ATP5MF/APP/COX6C/PPP1R1A/UQCRQ/WDR45B/NDUFB9/PDP1/NUPR1/HMGCS2/UQCRB/DLD/ATP5F1B/
           Count
GO:1901293   105
GO:0031647   121
GO:0048193    99
GO:0009060    93
GO:0015980   123
GO:0006091   136
```

Gene Ontology enrichment analysis revealed distinct pathway programs in CAF-rich and tumor-rich clusters. CAF-rich clusters were enriched for pathways related to extracellular matrix organization, extracellular structure organization, positive regulation of cell adhesion, leukocyte cell–cell adhesion, and regulation of T-cell activation and lymphocyte differentiation. In contrast, tumor-rich clusters were enriched for pathways associated with nucleoside phosphate biosynthesis, mitochondrial organization, aerobic respiration, macromolecule catabolic processes, regulation of protein stability, and energy metabolism. These differences indicate that CAF-rich regions are dominated by adhesion, matrix, and immune-regulatory pathways, whereas tumor-rich regions are dominated by metabolic and biosynthetic programs.

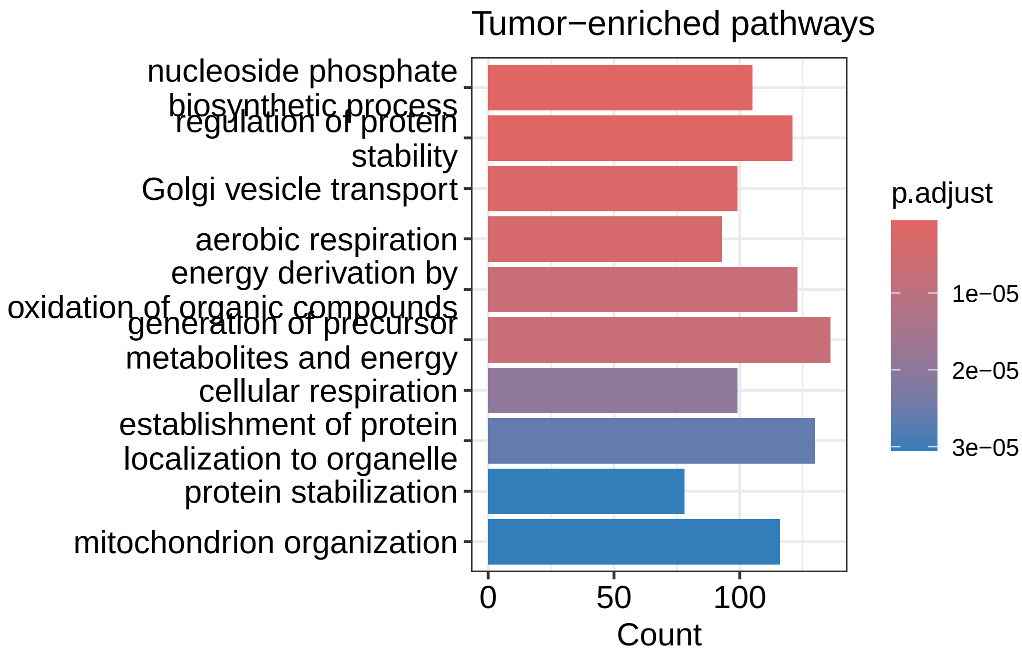
```
barplot(ego_caf, showCategory = 10, title = "CAF-enriched pathways")
```

```
Warning in (function (model, data, ...) : Arguments in `...` must be used.
x Problematic argument:
* by = "Count"
i Did you misspell an argument name?
```




```
barplot(ego_tumor, showCategory = 10, title = "Tumor-enriched pathways")
```

```
Warning in (function (model, data, ...) : Arguments in `...` must be used.
x Problematic argument:
* by = "Count"
i Did you misspell an argument name?
```



The pathway enrichment analysis further supports the presence of specialized tumor–stroma ecosystems. CAF-rich clusters showed strong enrichment of extracellular matrix organization and cell adhesion pathways, consistent with matrix-remodeling and physical barrier functions of activated CAFs (***Kalluri, Nat Rev Cancer 2016; Elyada et al., Cancer Cell 2019***). Enrichment of leukocyte adhesion, regulation of T-cell activation, and lymphocyte differentiation pathways suggests that CAF regions actively participate in immune regulation and immune exclusion, in line with reports linking LRRC15 CAFs and TGF- –driven stroma to immune checkpoint blockade resistance (***Dominguez et al., Cancer Cell 2020; Mariathasan et al., Nature 2018***). Conversely, tumor-rich clusters were enriched in mitochondrial organization, oxidative metabolism, nucleotide biosynthesis, and protein stability pathways, reflecting elevated metabolic and biosynthetic demands of proliferating tumor nests (***Pavlova & Thompson, Cell Metab 2016; Vander Heiden et al., Science 2009***). Together with the signature analysis, these pathway programs support a model in which metabolically active, COXIS-high tumor nests coexist with LRRC15 CAF-rich, matrix-remodeling stroma that modulates immune activation, forming a spatially organized microenvironment associated with treatment adaptation and immune evasion.

CONCLUSION

Our spatial transcriptomic analysis supports the hypothesis that COXIS-high tumor nests are spatially adjacent to LRRC15 CAF-rich stroma, forming organized tumor–stroma ecosystems in breast tumors. Dimensionality reduction, clustering, signature scoring, and pathway enrichment collectively showed that tumor-rich clusters were COXIS-high and metabolically active, while CAF-rich clusters were LRRC15-CAF–high, EMT-high, and enriched for extracellular matrix and immune-regulatory pathways.

LIMITATIONS

One of the limitations of our study is that it is based on a single spatial transcriptomics dataset and may not capture the full biological variability present across breast cancer patients and molecular subtypes.
Secondly, the spatial resolution of the platform means that each spot may contain a mixture of cell types, so regions classified as "tumor-rich" or "CAF-rich" are unlikely to be composed exclusively of those respective populations.
The other issue is that the analytical approaches used here i.e. gene-signature scoring and Gene Ontology enrichment are correlative; they reveal strong associations between COXIS-high tumor nests and LRRC15 CAF-rich stroma but cannot establish whether one process causes the other or directly drives immune-checkpoint blockade resistance.

FUTURE DIRECTIONS

1. Future studies should analyze larger cohorts and multiple tumor subtypes to test whether the same spatial ecosystem (COXIS-high tumor + LRRC15 CAF stroma) is consistently observed across patients.
2. Higher resolution approaches such as single-cell spatial transcriptomics, spatial proteomics, and multiplex imaging should be used to more precisely separate tumor cells, CAFs, immune cells, and vascular cells within each spot.
3. Functional experiments are needed to test causality, including:
   - co-culture of tumor cells with LRRC15 CAFs
   - inhibition of COX-2/COXIS or TGF- pathways
   - in vivo models testing immune infiltration and ICB response

   These approaches might be able to determine whether altering CAFs or COXIS programs changes immune exclusion or therapy resistance, moving beyond correlation.